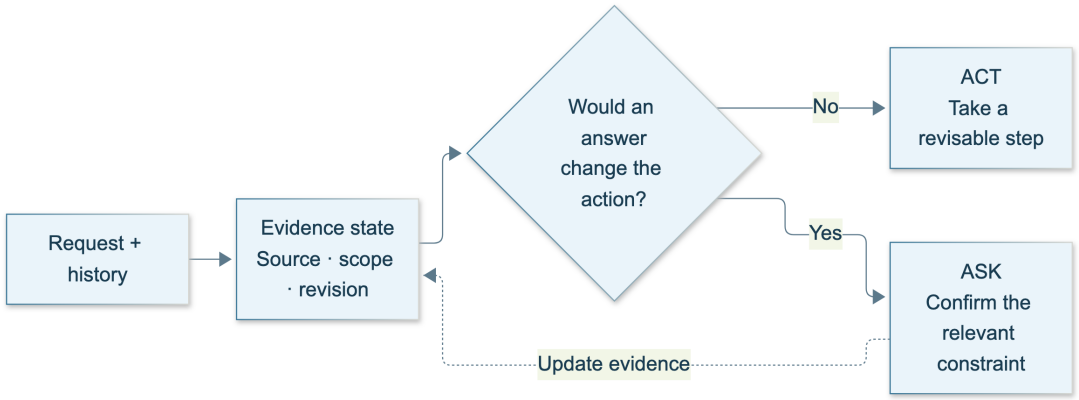
I3Agent: When to Ask, and What to Ask

One-page research sketch · 26 October 2026

Related work, research ideas, and experiment plans: I3Agent project on Notion.

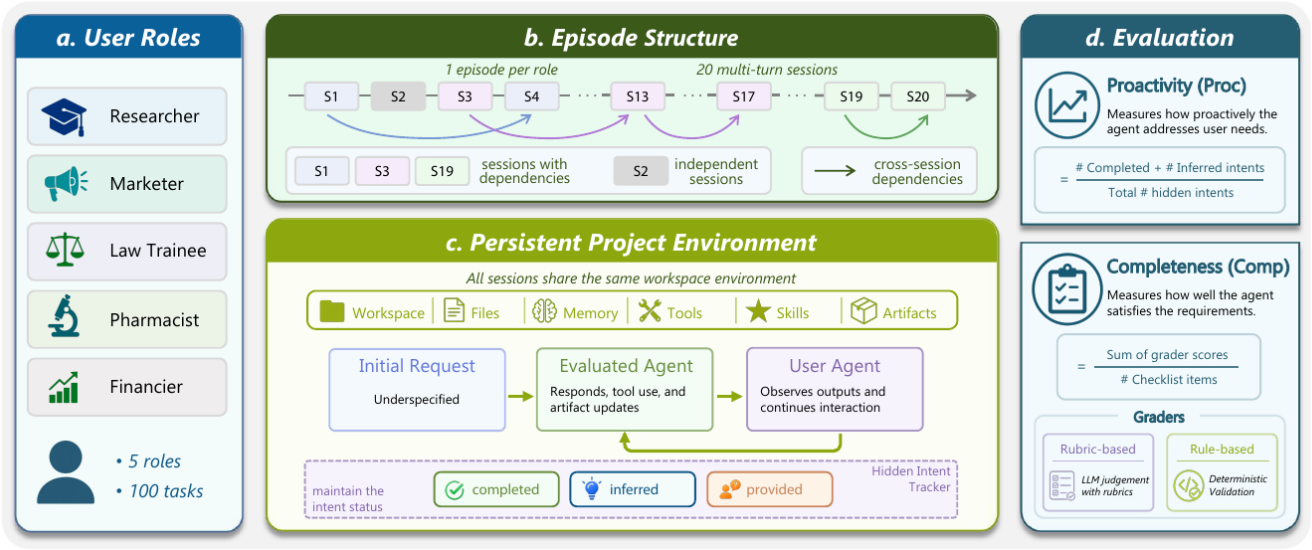
1. Narrow claim

Hypothesis: tracking whether past preferences are confirmed, scoped, or revised improves task success at the same question budget, compared with history retrieval and a memory-matched value-of-information (VoI) policy. Test the effect on question selection, beyond merely guarding actions.



Proposed controller. Example: a past $50 budget + a special occasion -> ask whether the cap still applies.

2. Closest work & starting environment



Original Figure 1, Pi-Bench paper: 100 tasks, persistent workspaces, hidden intents; code & data.

Decision baselines: VoI, REVOIR, Steerable Clarification. **Closest state representation:** Grounding Awareness. **Follow-ups:** LifeSim code for connected sessions; PAST-Bench for persistence controls.

3. Smallest controlled experiment

- **Setup:** reproduce one 20-task Pi-Bench sequence; construct 30 held-out checkpoints (10 each: sufficient evidence, missing fact, outdated/scoped preference).
- **Controls:** direct action, uncertainty threshold, memory-matched VoI, and our controller. Same model/history/tools; at most 1 atomic question. Scripted vs native users; three generation seeds.
- **Readout:** success, wrong assumptions, question/answer burden, and total compute. Pilot go/no-go: at least 10-point success gain over the strongest control at matched question count, no more consequential errors, consistent across user protocols. Report paired intervals.

4. Likely failure mode

Simulator hacking: native users volunteer unmet requirements. Audit disclosure and match compute. Give the true constraint as a diagnostic: persistent failure implicates execution. No training in the pilot; self-adaptation comes later.